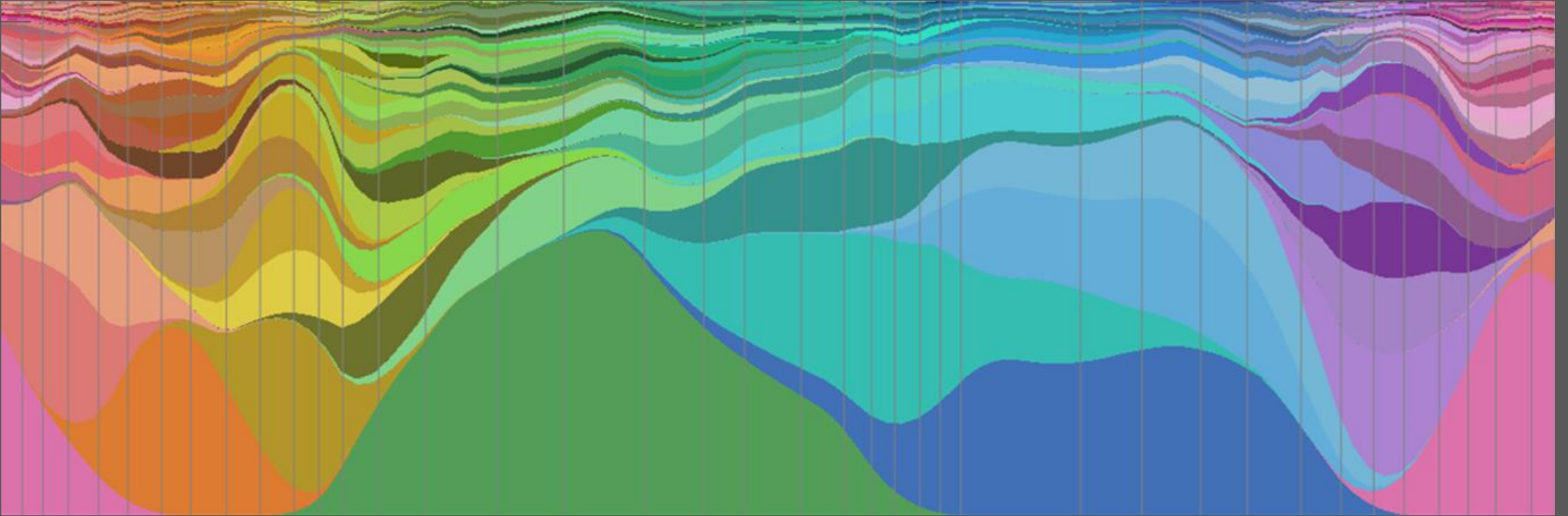


LDSCI5209 - Information Presentation and Visualisation

Week 9: Networks and Trees



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Recap: Maps and scientific visualisations

- Maps
 - Types
 - Projections
 - Resources
- Scientific visualisations
 - Uncertainty
 - Confidence intervals
 - Density

Plan for today

- What is a network
- What is a tree
- Common visual encoding techniques for network data

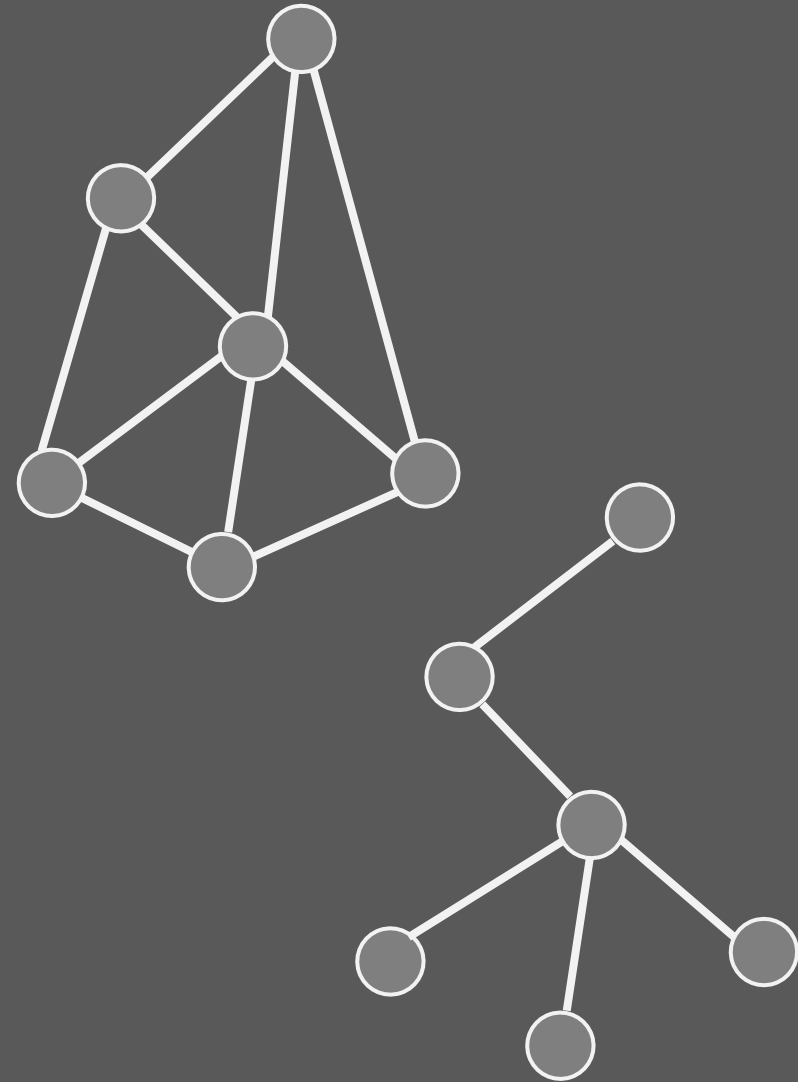
Networks and trees

- **Networks (Graphs):**

- Models relations among data
- Nodes and edges

- **Trees:**

- Graphs with hierarchical structure
- Connected graph with $N-1$ edges
- Nodes as parents and children



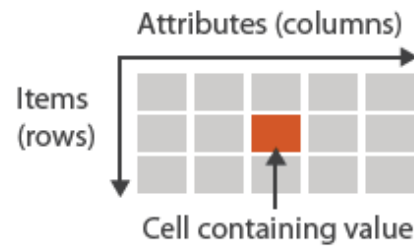
Networks

Relationships, nodes, edges, directed/undirected, weights, applications

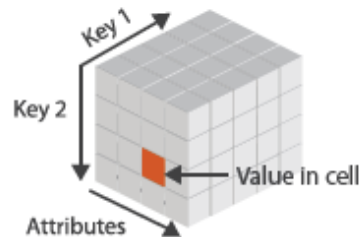
Network data types

➔ Dataset Types

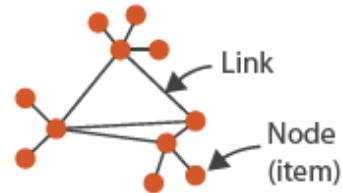
➔ Tables



➔ Multidimensional Table



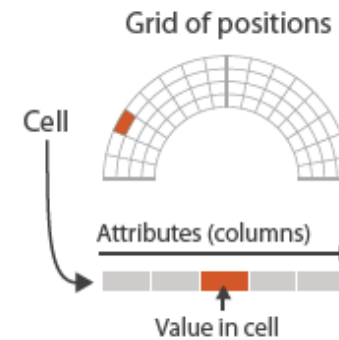
➔ Networks



➔ Trees



➔ Fields (Continuous)



➔ Geometry (Spatial)



Arrange Spatial Data

➔ Use Given

➔ Geometry

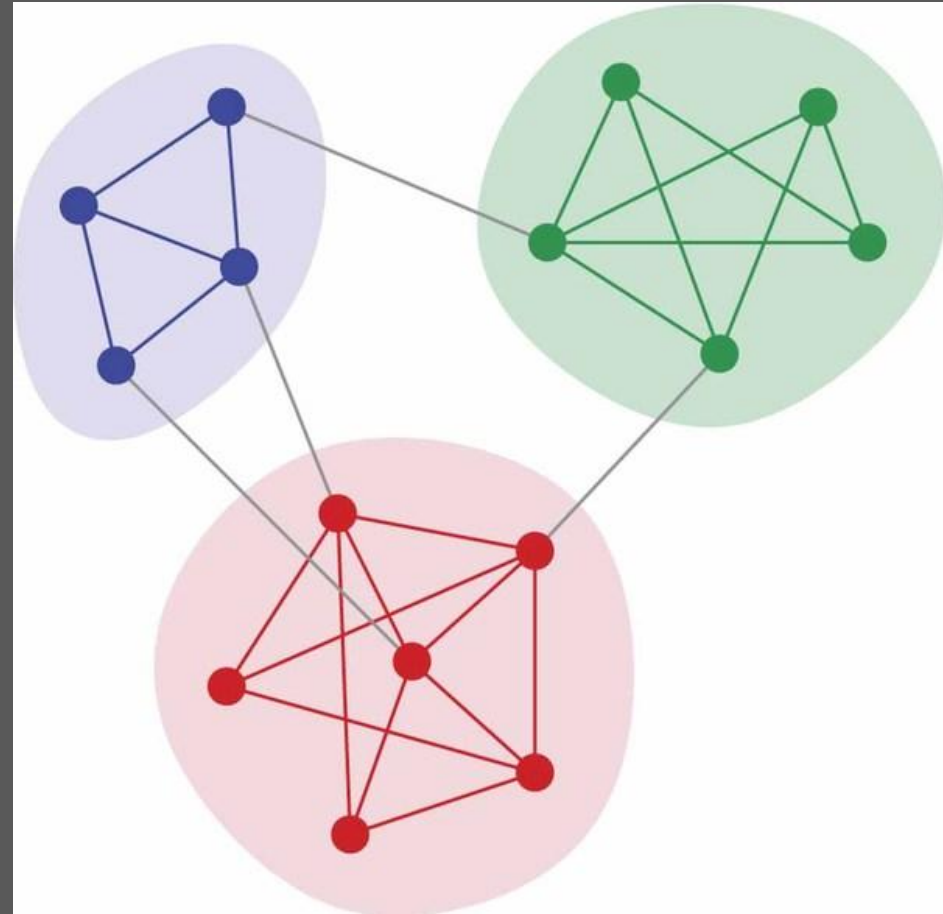
➔ Geographic

➔ Other Derived



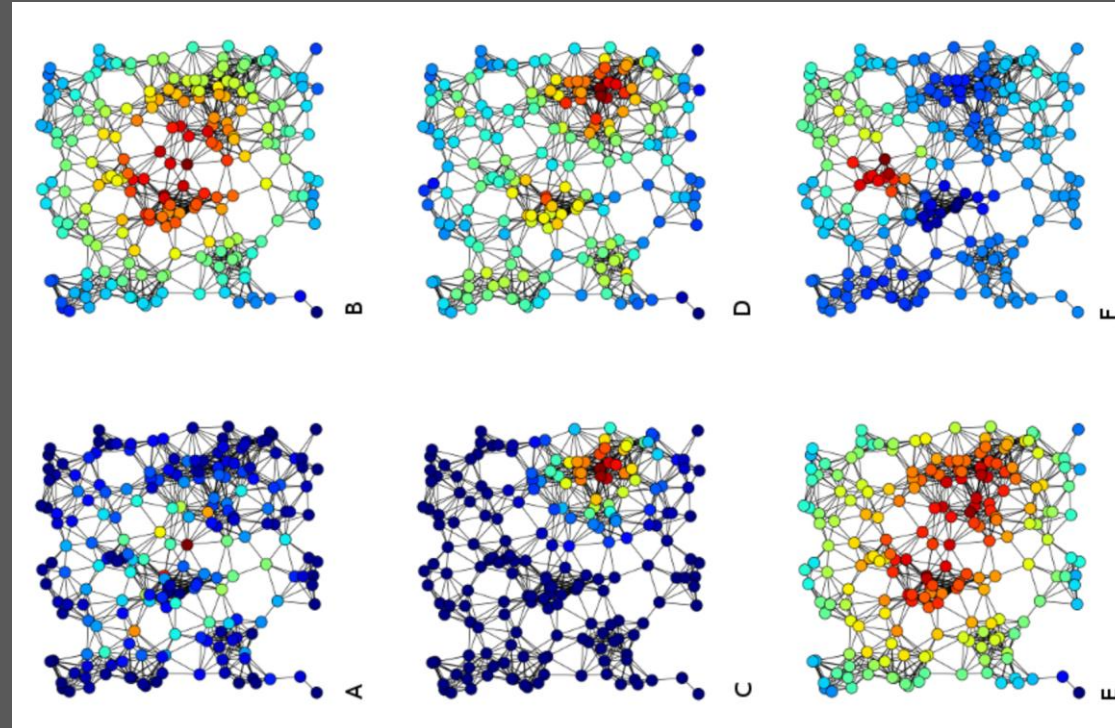
Terminology

- graph = network
- nodes = vertices, actors
- edges = links, relations
- clusters = communities



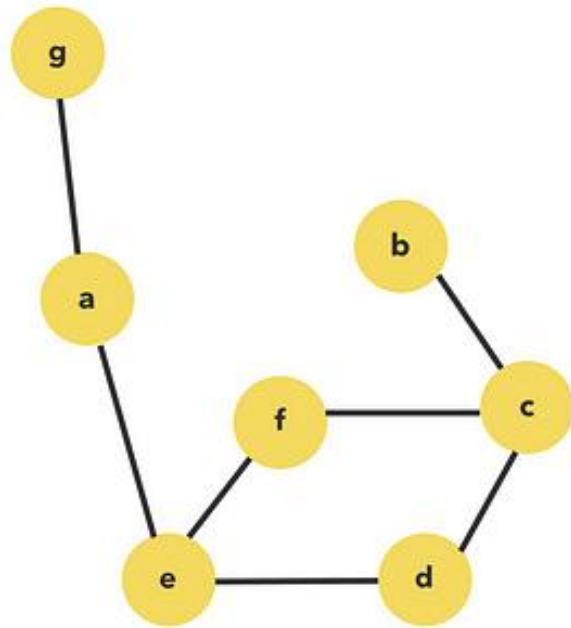
Centrality

Importance of *nodes* within a graph

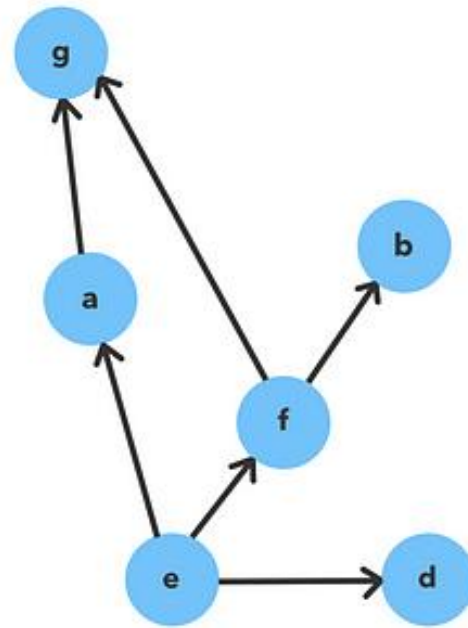


Edge types

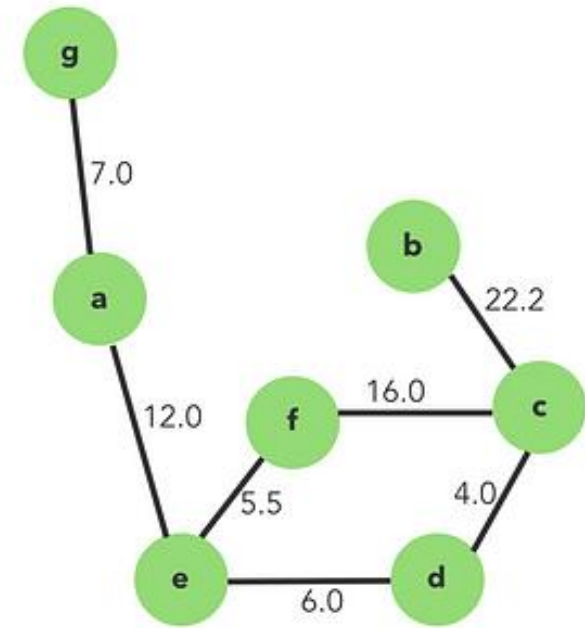
Undirected



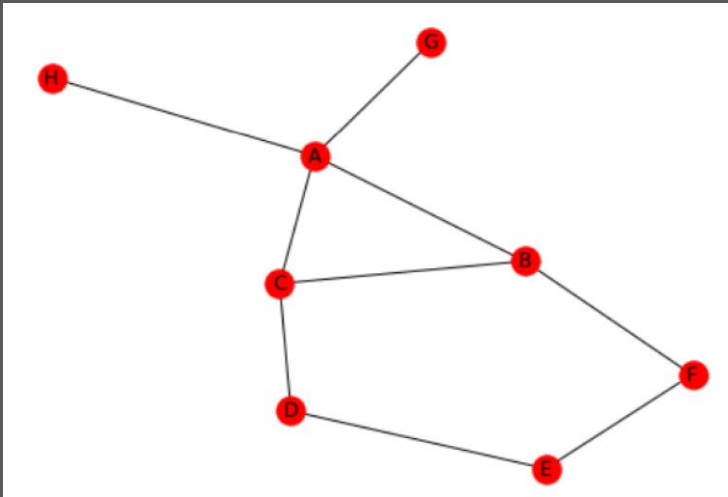
Directed



Weighted



Adjacency



Adjacency Matrix

	A	B	C	D	E	F	G	H
A	0	1	1	0	0	0	1	1
B	1	0	1	0	0	1	0	0
C	1	1	0	1	0	0	0	0
D	0	0	1	0	1	0	0	0
E	0	0	0	1	0	1	0	0
F	0	1	0	0	1	0	0	0
G	1	0	0	0	0	0	0	0
H	1	0	0	0	0	0	0	0

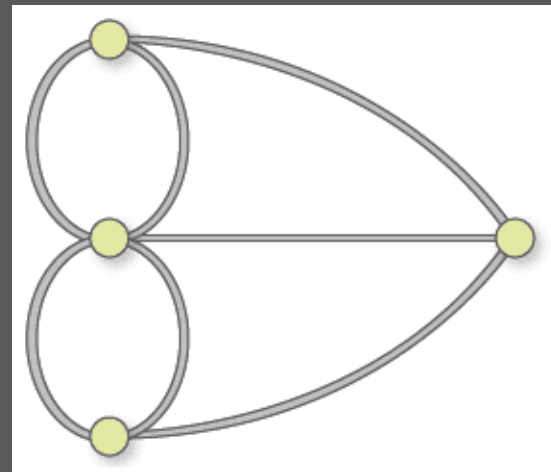
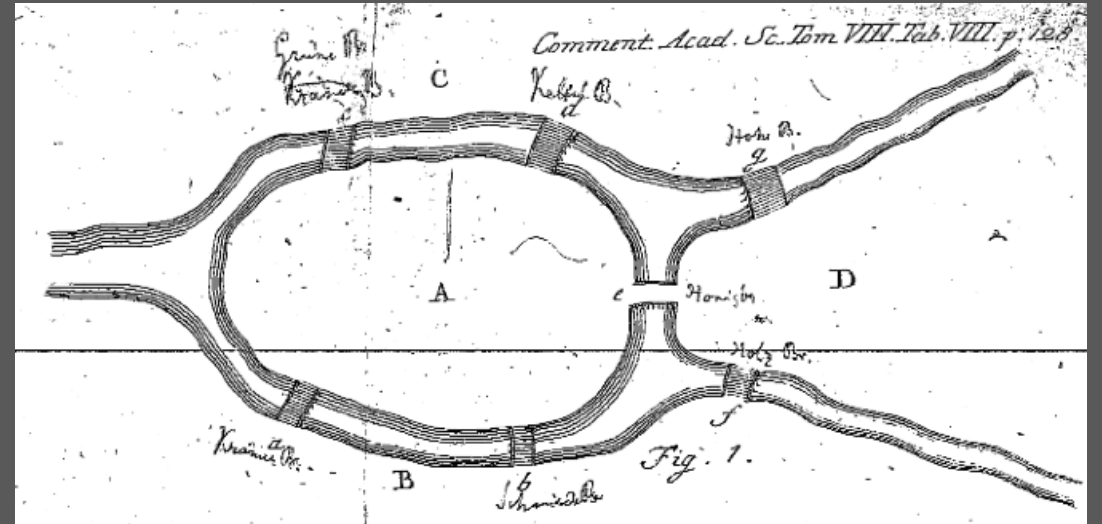
Adjacency List

[A] => [C, G, H, B]
[B] => [C, A, F]
[C] => [A, D, B]
[D] => [C, E]
[E] => [D, F]
[F] => [B, E]
[G] => [A]
[H] => [A]

Paths in graphs

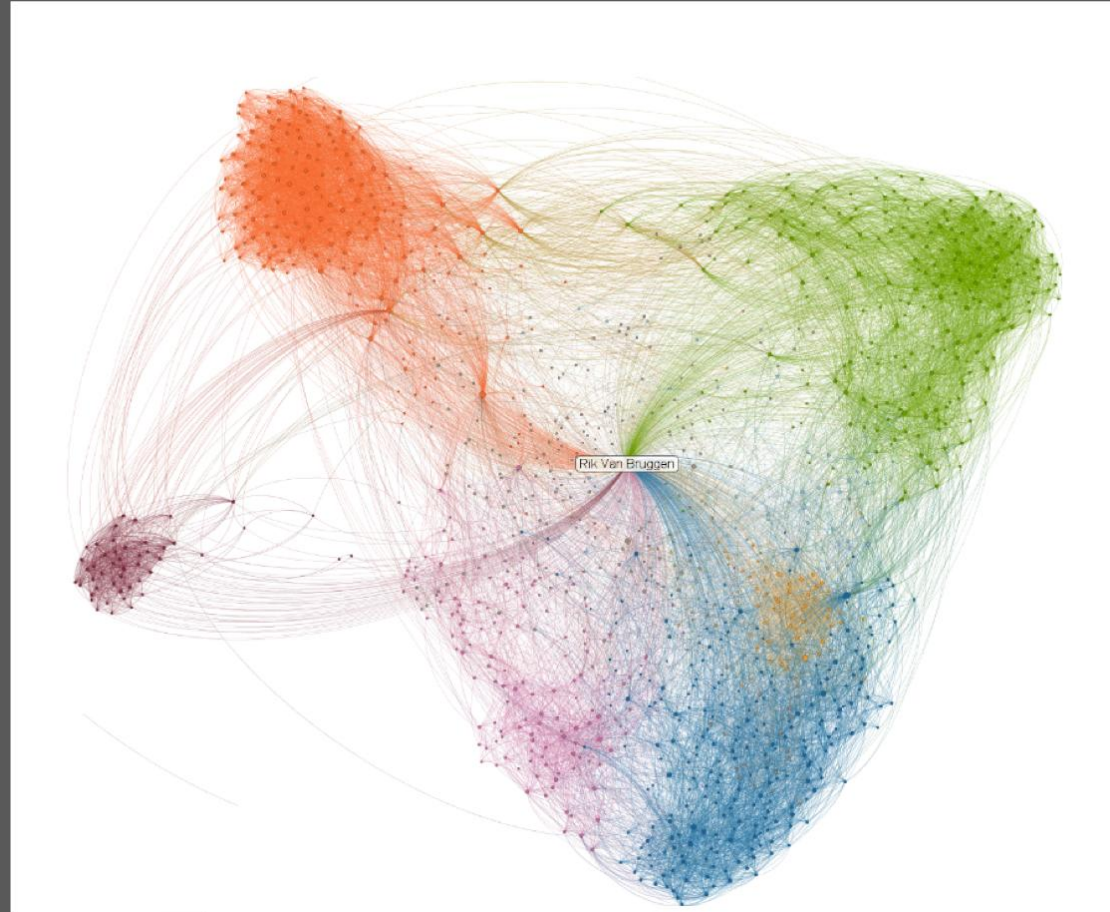
Euler path is a path that uses every edge of a graph exactly once.

A graph G has a Euler path if and only if there are exactly two vertices with odd degree.



Leonhard Euler (1707 – 1783)

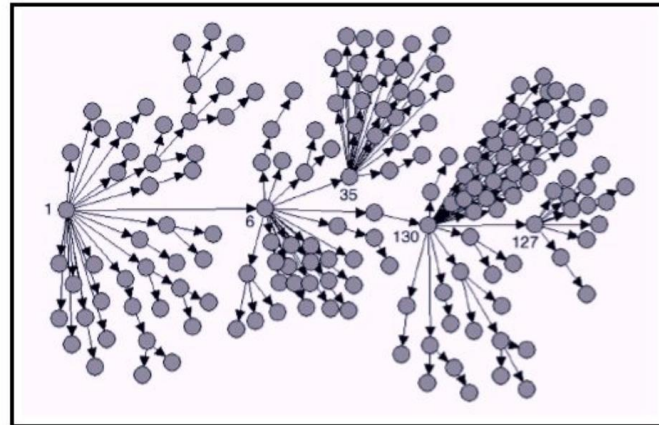
Networks in social media



Networks in pandemics

Is there a small subset of SARS patients who account for a disproportionate share of transmission? MMWR May 9, 2003 / Vol. 52 / No. 18 - I

FIGURE 2. Probable cases of severe acute respiratory syndrome, by reported source of infection* — Singapore, February 25–April 30, 2003

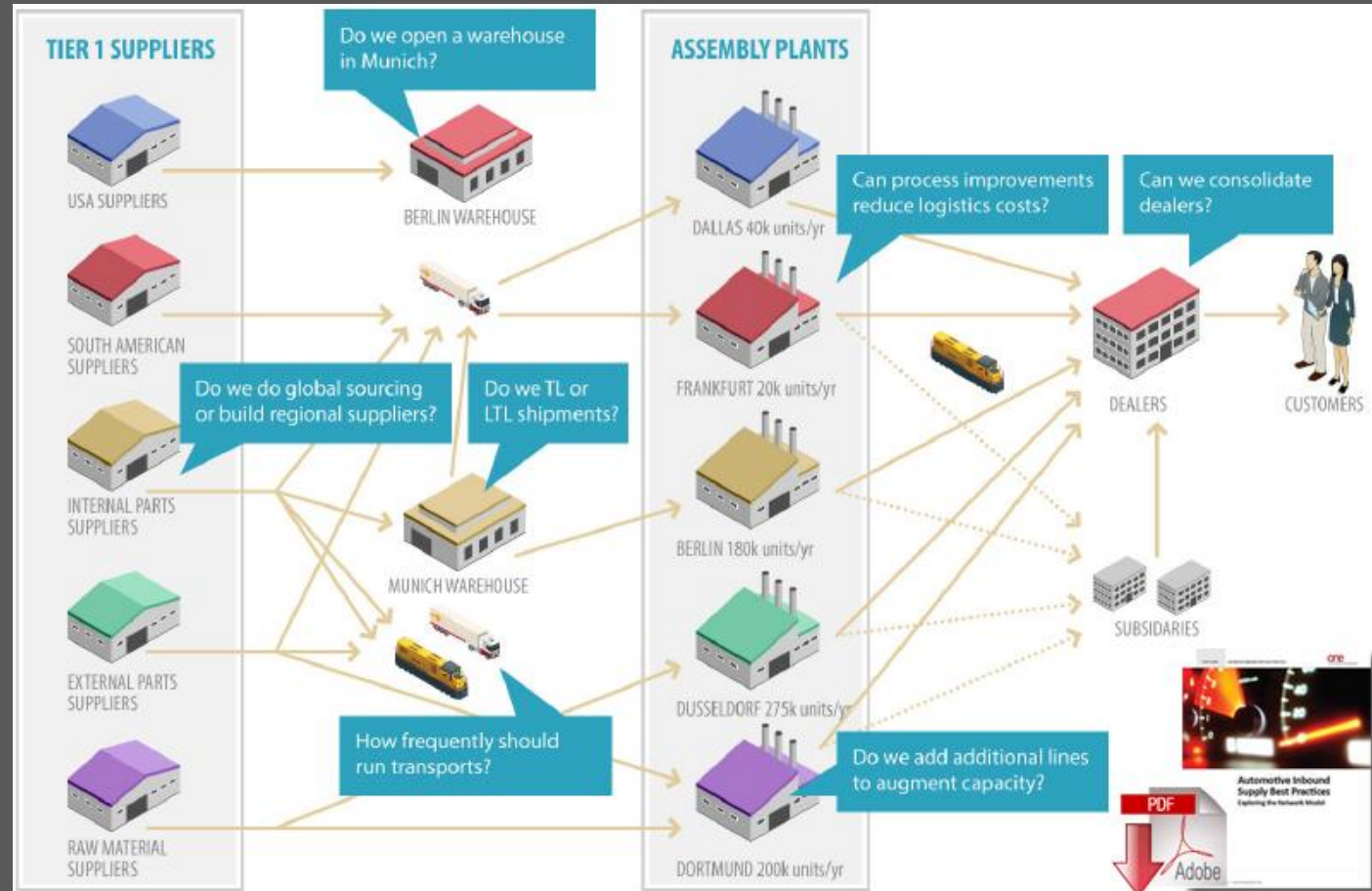


* Patient 1 represents Case 1; Patient 6, Case 2; Patient 35, Case 3; Patient 130, Case 4; and Patient 127, Case 5. Excludes 22 cases with either no or poorly defined direct contacts or who were cases translocated to Singapore and the seven contacts of one of these cases.

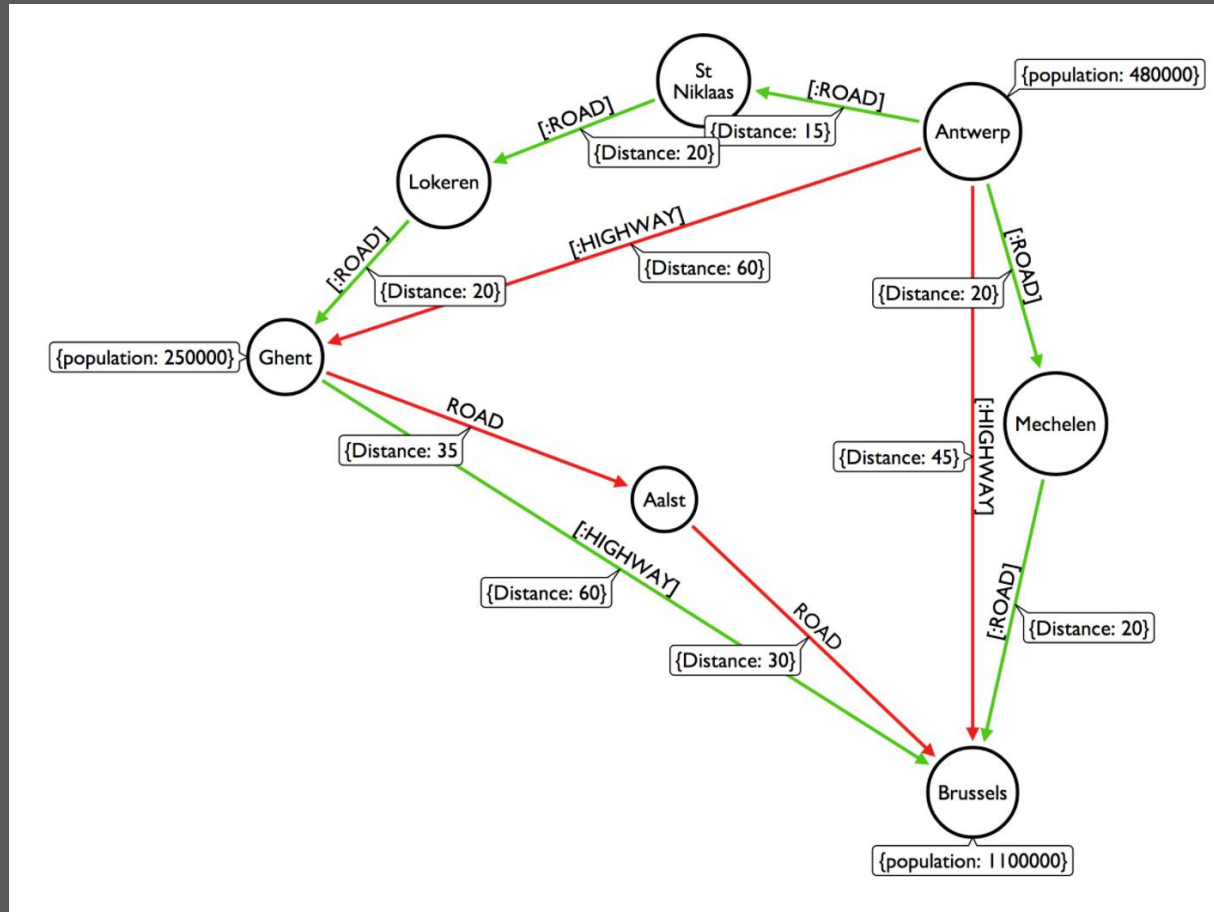
Reference: Bogatti SP. Netdraw 1.0 Network Visualization Software. Harvard, Massachusetts: Analytic Technologies, 2002.

Patients No: 1, 6, 35, 130&127 seemed to be “hypertransmitters”

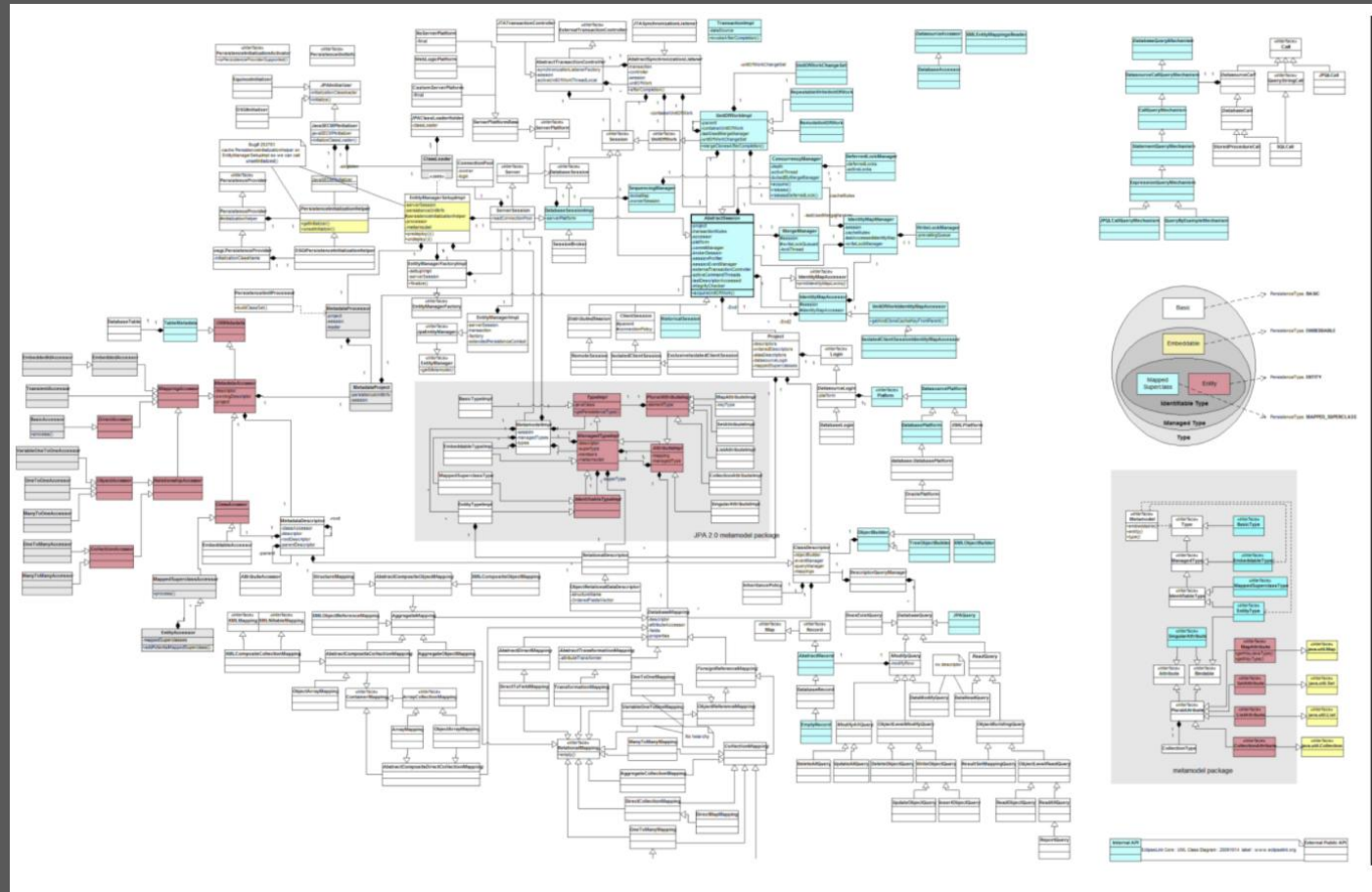
Networks in logistics



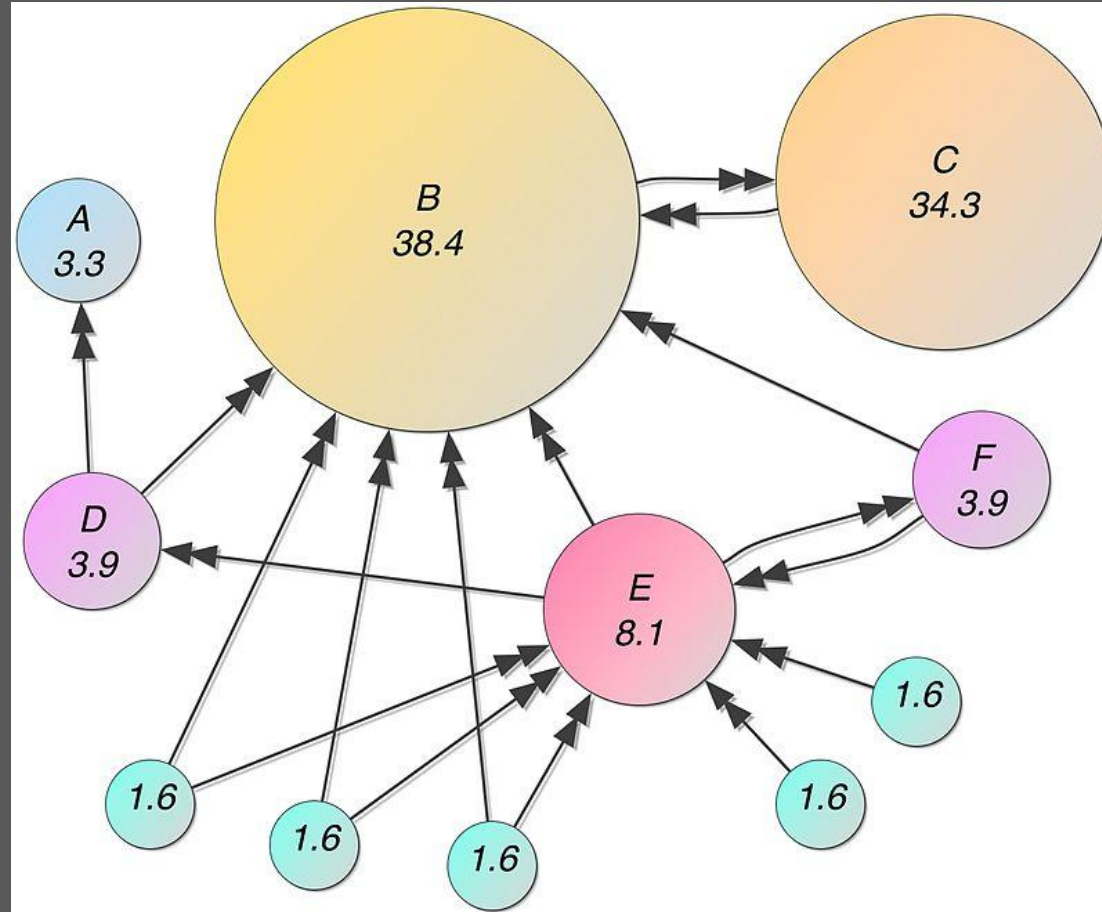
Networks in routing



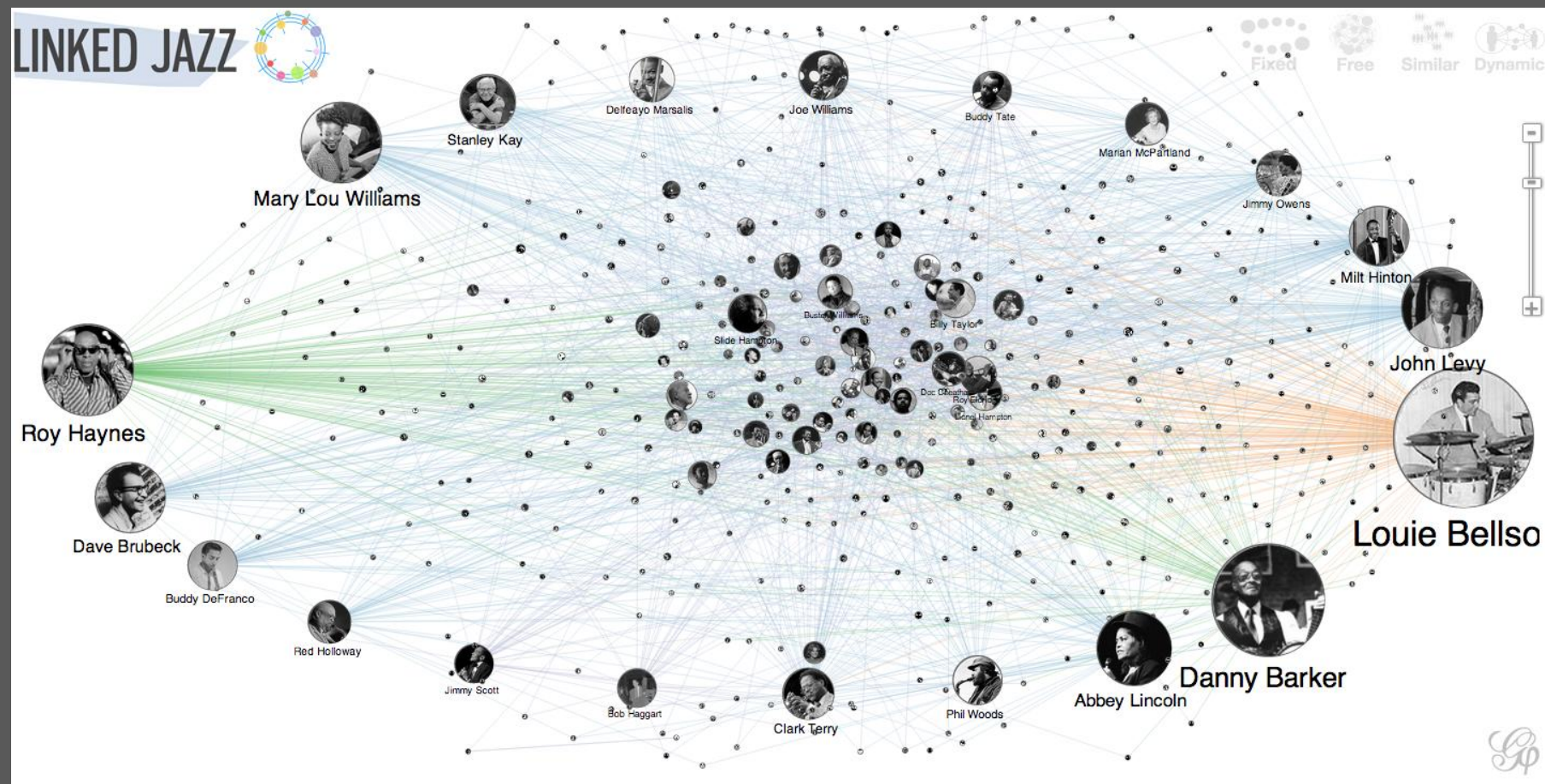
Networks in software engineering



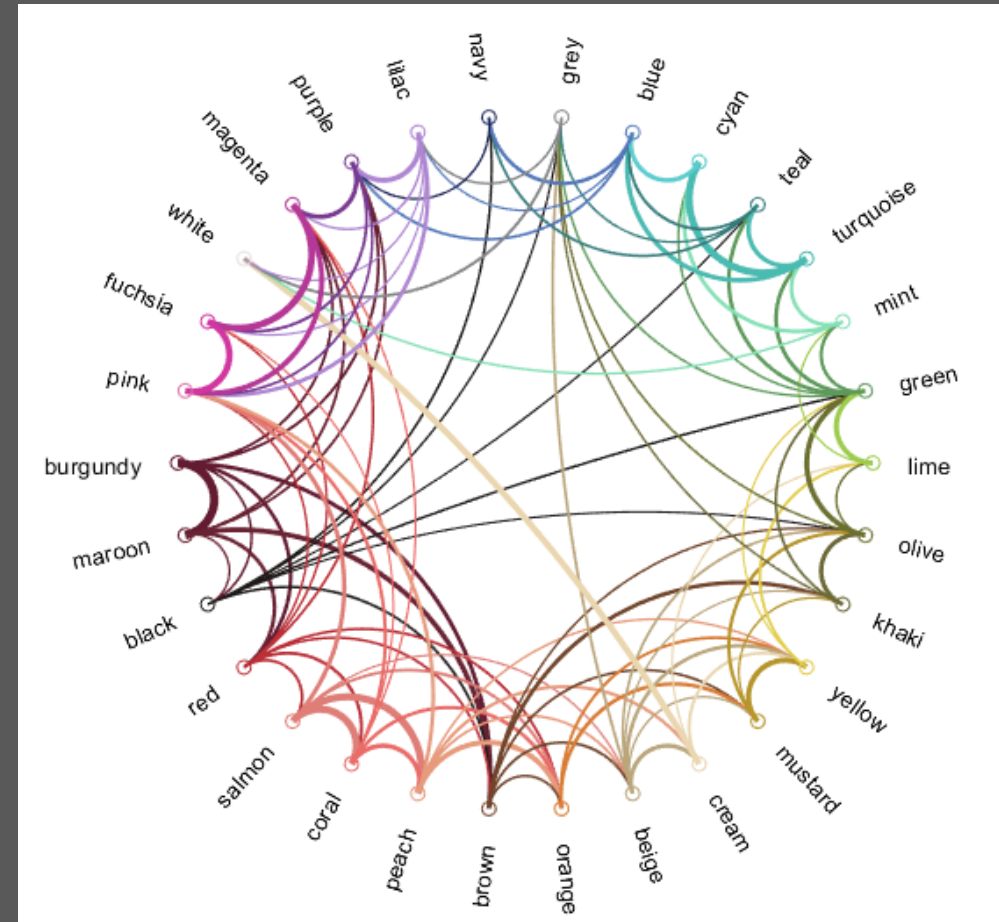
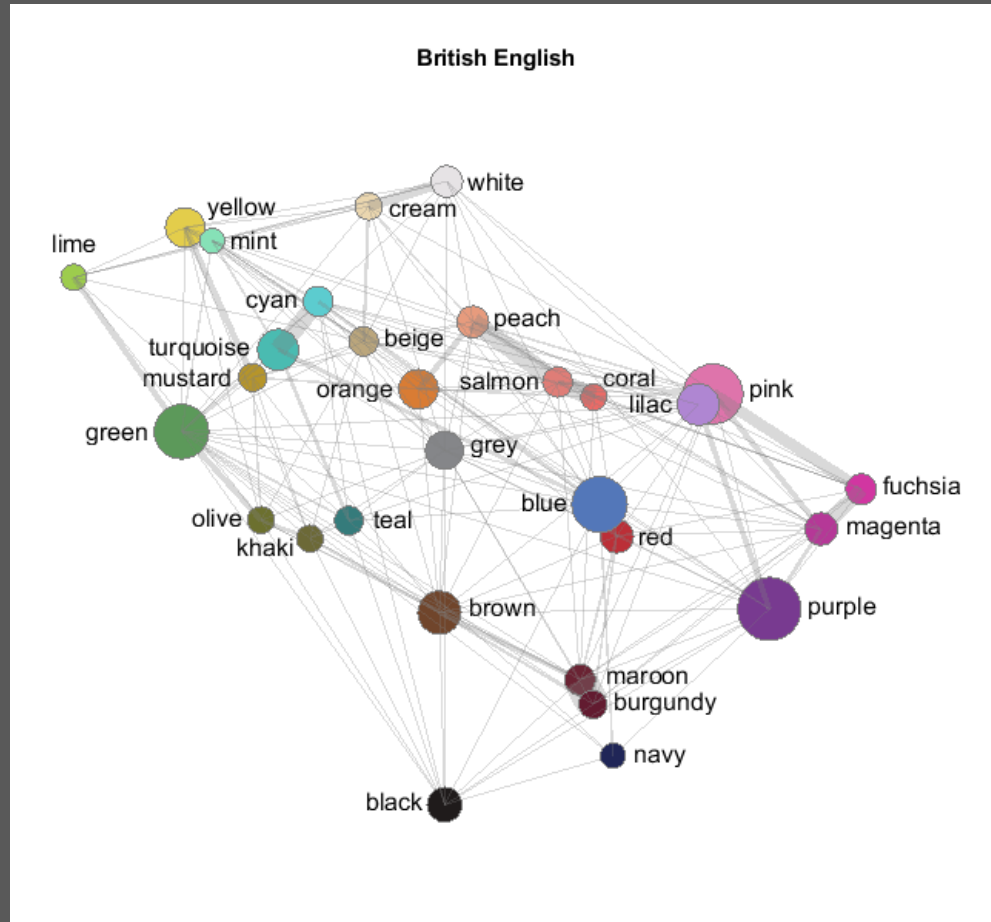
Networks in search engines



Networks in music analysis



Networks in colour lexicons

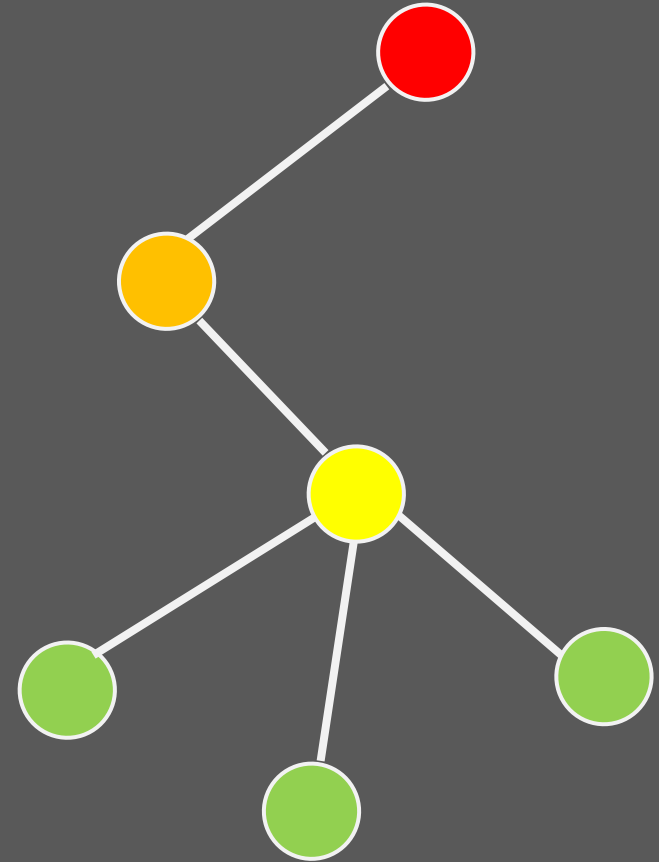


Trees

Components, types, applications, challenges

Terminology

- Root = starting point
- Child = descendant node of a node
- Parent = predecessor of a node
- Siblings: children with the same parent
- Leaf = a node with no children



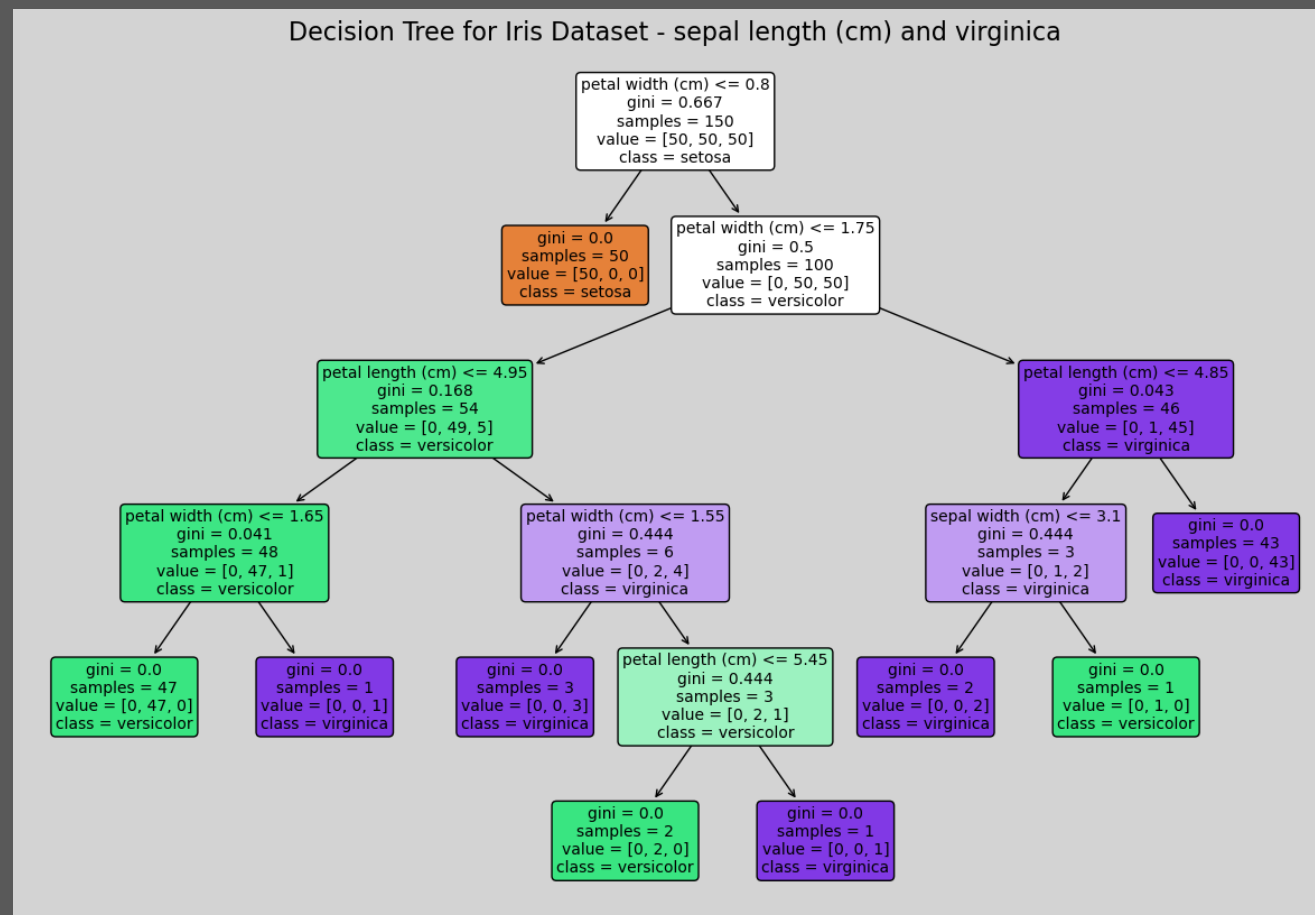
Why use trees?

- **Hierarchical Representation:** Ideal for representing data with parent-child relationships.
- **Clarity:** Helps understand complex structures in a simplified manner.
- **Interactive Exploration:** Allows users to zoom in and out of various levels.

Types of trees

- **Decision Trees:** Used in classification and regression tasks.
- **Hierarchical Trees:** Used to show the relationships between categories or objects.
- **Binary Trees:** Each node has at most two children, often used in searching algorithms.

Decision, hierarchical and binary *iris* tree



Challenges of graph visualisations

- **Overcrowding:** Large trees can become difficult to read.
- **Depth vs. Breadth:** Managing too many levels or too many nodes at each level.
- **Interactivity:** Ensuring users can easily navigate complex tree structures.

Questions?